Aditya Reddy

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Education

Manipal University Jaipur **Aug 2024 – May 2028**

Bachelor of Technology, Electronics and Communication Engineering *Jaipur, India*

- **Relevant Coursework:** Digital Logic Design, Circuit Analysis, Signals & Systems, Electronic Devices, Linear Integrated Circuits

Technical Skills

HDL / RTL Design: SystemVerilog, Verilog; clock-domain crossing, SPI/CSR interfaces, FIFO and memory-controller design

Physical Design: RTL-to-GDSII, floorplanning, hard-macro integration, PDN, place & route, CTS, multi-corner STA, SDC authoring, DRC/LVS/antenna signoff

EDA & Tools: LibreLane, OpenLane 2, OpenROAD, OpenSTA, Yosys, OpenRAM, Magic, KLayout, Vivado, SymbiYosys, Verilator, sv2v, GTKWave, LTspice; SKY130 PDK

Programming & Systems: C, C++, Python; Linux, Git, Make, GCC, RISC-V toolchain, lock-free concurrency, FIX protocol, TCP/IP, POSIX

Projects

INT8 Matrix-Multiply Accelerator – RTL to GDSII on SKY130 | SystemVerilog, LibreLane **GitHub**

- Designed a weight-stationary 4×4 INT8 systolic matrix-multiply accelerator in SystemVerilog — pipelined MAC array, SPI slave control interface with clock-domain crossing, CSR/IRQ block, and an SRAM-backed input queue — verified to 194/194 directed tests and gate-level simulation against the synthesised netlist.
- Recovered 18 ns of setup slack on the reg-to-reg critical path by pipelining a 381-load control broadcast into per-column registered waves and removing 256 redundant operand flops, cutting worst-net fanout 5× and worst slew from 15.4 ns to 1.4 ns.
- Closed timing at 12 ns (83 MHz) and signed off clean on post-route STA at all nine PVT corners — zero setup, hold, DRC, LVS, antenna and PDN violations — by root-causing the critical path to an SRAM macro read arc that launched on the falling edge and re-clocking the macro read port; 74.2% utilisation, 15.5 K cells and one macro on a 0.48 mm^2 die.

RISC-V Processor Core – RV32IM 5-Stage Pipeline | SystemVerilog, Vivado, Verilator **GitHub**

- Implemented a 5-stage RV32IM pipeline with EX/MEM forwarding, load-use stall insertion and 28 SVA assertions at 100% pass rate across 8 test programs; added single-cycle 64-bit multiply and a 32-cycle non-restoring radix-2 divider that stalls the pipeline via a `ready` signal, covering all M-extension edge cases.
- Built a 3-component dynamic branch predictor (16-entry BTB, 256-entry Gshare PHT, 4-deep RAS) to cut conditional branch penalties, then synthesised to Artix-7 in Vivado, closing timing at 100 MHz with performance counters quantifying IPC, branch mispredict rate, and stall cycles.

Dual-Clock Async FIFO – Formally Verified CDC Design | SymbiYosys, OpenRAM, SKY130 **GitHub**

- Designed a parametric dual-clock FIFO with gray-coded pointers, 2-flop synchronisers and an OpenRAM pseudo-dual-port SRAM (16×64) as storage; formally proved no-overflow, no-underflow and CDC safety with SymbiYosys k-induction (`mode prove`) across isolated `.sby` targets per sub-module.

Low-Latency Trading Gateway | C++, FIX Protocol, Lock-Free Concurrency **GitHub**

- Built a dual-thread market-data and order-processing pipeline using Boost SPSC lock-free queues and a pre-allocated memory pool, pinning threads with `SCHED_FIFO` to eliminate mutex contention; implemented a zero-copy single-pass FIX parser with `std::from_chars`, achieving 1.4 μs average parse latency and 2.8 μs minimum tick-to-trade.

Activities

Volunteer — Bison Asha School for Special Needs: supported educational activities and organised inclusive events 2022–2024